



Cell Injury

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Pathology 201



Causes of cell injury

Hypoxia

Physical agents

Chemical agents and drugs

Infectious agents

Immunologic reactions

Genetic derangements

Nutritional imbalances



Hypoxia

Hypoxia is the most common cause of cell injury.

Ischaemia is the most common cause of hypoxia.

Ischaemia injures tissue faster than hypoxia alone because substrate delivery also stops.



ATP depletion

ATP depletion causes failure of the Na⁺/K⁺ ATPase pump.

Failure of the sodium pump leads to influx of sodium and water, producing cellular swelling.

Reduced ATP also causes a switch to anaerobic glycolysis, which lowers intracellular pH.

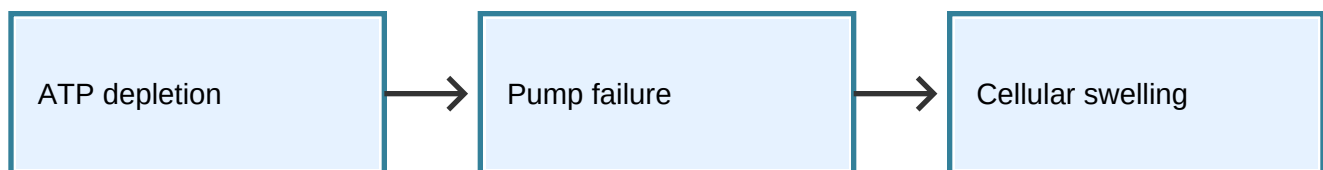


Figure 4.1: from ATP depletion to cellular swelling



Cellular swelling

Cellular swelling is the first manifestation of almost all forms of injury to cells.

Cellular swelling is caused by failure of energy-dependent ion pumps in the plasma membrane.

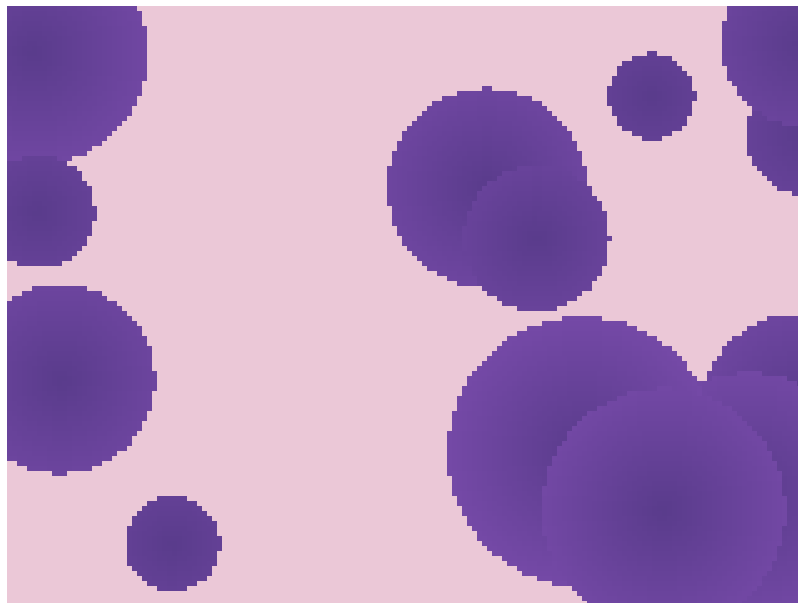


Figure 5.1: hydropic change of renal tubular cells



Fatty change

Fatty change (steatosis) is the accumulation of triglycerides within parenchymal cells. Fatty change is seen in hypoxic, toxic and metabolic injury, mainly in the liver.

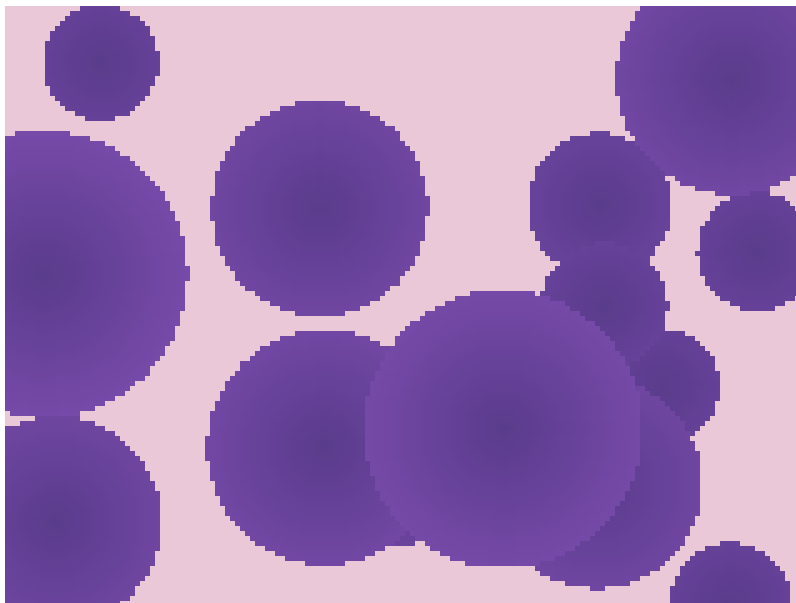


Figure 6.1: steatosis of hepatocytes



Reversible versus irreversible injury

Reversible injury shows cellular swelling and fatty change.

Irreversible injury is defined by severe membrane damage and mitochondrial dysfunction.

Membrane damage marks the transition to irreversible injury.



Patterns of necrosis

Coagulative necrosis

Liquefactive necrosis

Caseous necrosis

Fat necrosis

Fibrinoid necrosis

Gangrenous necrosis



Apoptosis

Apoptosis is programmed cell death mediated by caspases.

Apoptosis does not elicit an inflammatory reaction.

Cytochrome c release from mitochondria activates caspase-9 in the intrinsic pathway.



References

Robbins and Cotran Pathologic Basis of Disease, 10th edition.

Thank you

Questions?